

Sequencing Peace: How Patterns of Cooperation Shape Public Opinion on Rapprochement

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Abstract

How does the trajectory of cooperation from a foreign adversary shape public support for peace? Existing research conceptualizes cooperative signals observed through rapprochement as isolated events, yet citizens often assess an adversary's behavior retrospectively and use these temporal records to infer future intentions. We argue that these cooperation trajectories matter because, even when rapprochement consists of positive interactions overall, negative changes in cooperation can erode public evaluations by resonating with prior beliefs that the adversary is inherently hostile. Thus, citizens exposed to temporal patterns in which cooperation deteriorates or alternates with decline form lower expectations of future cooperation and express stronger resistance to peace initiatives. We test this argument using two survey experiments: a hypothetical U.S.–China rapprochement scenario in the South China Sea ($N = 2,324$) and a selective-history experiment examining the 2018 inter-Korean rapprochement in South Korea ($N = 2,000$). Across both studies, respondents exposed to inconsistent or descending cooperation trajectories anticipated lower levels of future adversarial cooperation and showed less support for conciliatory policies than those exposed to alternative trajectories or cooperation without a discernible pattern. These findings explain why rapprochement that features exchanges of meaningful concessions often fails to secure the public endorsement necessary for durable peace.

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1. Introduction

Peace initiatives often falter not because leaders fail to reach across a hostile divide, but because publics remain unconvinced that the adversary’s cooperation is genuine or durable. Rapprochement — the initial phase in which state leaders seek to reduce hostility and establish trust with a foreign adversary — therefore requires not only diplomatic leadership but also public endorsement. Popular support helps leaders sustain reconciliation efforts against hawkish rivals domestically (Vasquez, 1993; Lebow, 1997; Colaresi, 2004) and send credible signals of benign intentions to their international counterpart (Brandt et al., 2008). More fundamentally, only when citizens embrace cooperation can elite-level rapprochement develop into the societal conciliation that defines enduring “warm peace” (Ripsman, 2016; Diehl et al., 2021; Mattes & Weeks, 2024).

Existing research explains such support primarily through the adversary’s olive branch. One line of work shows that reciprocated cooperation encourages citizens to endorse conciliatory policies and bolsters dovish domestic leadership (Colaresi, 2004; Kupchan, 2010; Davies & Johns, 2016). Another argues that costliness is decisive: because costly signals are harder to fake (Schelling, 1960; Fearon, 1997), bold cooperative gestures can convey benign intentions even under high distrust (Kydd, 2005). Recent empirical work supports this logic, showing that costly signals can shift perceptions of adversarial intentions and increase support for conciliatory policy (Kertzer et al., 2020; Yoder & Haynes, 2021; S. J. Kim, 2021; Mattes & Weeks, 2022).

However, empirical cases suggest that, even when leaders succeed in rapprochement by securing meaningful concessions from a foreign adversary, public support for peace has often proven shallow or only short-lived. For instance, despite the exchange of cooperative gestures during the Davos Process between Turkey and Greece in the late 1980s, deep societal animosity persisted: only a third of Greek citizens supported cooperative relations with Turkey, and less than half of the Turkish public endorsed even partial troop removal from Cyprus (Pridham, 1991; Öniş & Yilmaz, 2008). In the U.S., a week after President Obama announced Iran’s agreement to the 2015 nuclear deal, only 33 percent of the public approved of it — a figure that fell further to 21 percent within six weeks (Pew Research Center, 2015). More recently,

South Korean approval of President Moon Jae-in's conciliatory policy toward North Korea fell from 65 to 38 percent within two years ([B. Kim et al., 2021](#)), despite three historic inter-Korean summits in 2018 and the significant border demilitarization that followed. Broader historical evidence suggests indirect support for this pattern: of 289 rivalry dyads that experienced successful rapprochement between 1900 and 2020, only 60 advanced to societal-level conciliation ([Diehl et al., 2021](#)).

This raises a puzzle: why does public endorsement of peace initiatives often prove fragile, even when adversaries extend meaningful cooperative gestures? We posit that part of the answer lies in a dimension the existing literature has largely overlooked: the temporal structure of cooperation during rapprochement. Existing research treats exchanged cooperative signals as discrete events, evaluating their effects in isolation rather than as part of an unfolding sequence. Yet rapprochement is not a singular diplomatic act but a sustained process of intergovernmental exchanges ([Kupchan, 2010](#); [Mattes & Weeks, 2024](#)), and thus citizens rarely encounter cooperative signals in isolation. Media framing often situates new cooperative developments within the context of prior events, encouraging audiences to interpret each signal retrospectively as part of a broader trajectory. Moreover, a single gesture is unlikely to change citizens' deeply rooted hostile beliefs; rather, they tend to suspend judgment until repeated actions establish a clear behavioral pattern. Thus, it is through this cumulative interpretation of an evolving sequence that citizens form expectations about whether the adversary's cooperative behavior will endure and, consequently, whether conciliatory policy deserves their support.

We argue that cooperation trajectories shape citizens' forward-looking expectations by assigning disproportionate weight to negative changes. Citizens conditioned by longstanding enmity bring deeply entrenched prior beliefs that the adversary is untrustworthy and conflict is the natural state of the relationship ([Jervis, 1976](#); [Thompson, 2001](#)), making them more likely to assimilate incoming signals into existing hostile priors than to revise them ([Kertzer et al., 2020](#); [Han, 2025](#)). Consequently, even when absolute levels of cooperation remain positive throughout rapprochement, downward changes between periods resonate with hostile priors and carry disproportionate weight in citizens' evaluations, suppressing forward-looking optimism about future reciprocity.

Building on this, we identify two specific patterns that particularly dampen the effect of adversary cooperation on such expectations. First, an inconsistent pattern — in which cooperative signals alternate between increases and decreases — yields lower expectations, because declining cooperation assimilates more easily into hostile priors and is not equally offset by intervening periods of increase. Second, a descending trajectory — in which reductions in cooperation accumulate toward the least cooperative endpoint — continuously reinforces the prior image of the adversary as fundamentally hostile, depressing expectations of future cooperation. Both trajectories thus suppress public support for conciliatory policy relative to alternative patterns, such as consistent or ascending cooperation.

To test these arguments, we conduct two preregistered survey experiments in the U.S. (Study I) and South Korea (Study II). Study I presents American respondents ($N = 2,324$) with a hypothetical scenario of U.S.–China tensions in the South China Sea, in which China reduces its artillery facilities according to one of four cooperation trajectories (consistent, inconsistent, ascending, descending) or without any temporal pattern, holding the aggregate level of cooperation constant across conditions. This design allows us to isolate the causal effect of trajectory shape on public expectations and policy preferences, free from confounds introduced by real-world political context. Study II addresses the ecological validity limitations of the first by employing a selective-history design with South Korean respondents ($N = 2,000$). Rather than presenting an abstract scenario, this design exposes respondents to alternative framings of North Korea’s cooperative behavior during the 2018 inter-Korean rapprochement — a case in which respondents have direct experiential familiarity with the events in question.

Across both studies, cooperation trajectories shape public expectations and policy preferences beyond the mere presence of cooperative signals. In Study I, descending cooperation reduces expectations of future Chinese cooperation by approximately 43 percent and support for withdrawing U.S. military presence by roughly 67 percent relative to the ascending condition; inconsistent cooperation similarly suppresses both outcomes, with mean policy approval statistically indistinguishable from zero. These effects persist even relative to the patternless cooperation baseline. Study II replicates these findings in a more realistic context. The same directional pattern is most evident in South Koreans’ expectations of North Korea’s future mil-

itary disengagement and diplomatic cooperation, with support for conciliatory policies moving in the expected direction. These results demonstrate that the temporal unfolding of cooperation is a consequential determinant of public support for rapprochement, with effects driven primarily by the suppressive influence of declining cooperation.

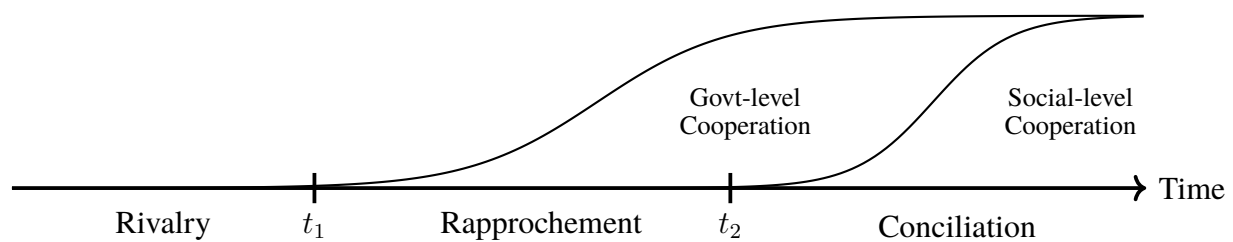
Our findings speak to two broader bodies of IR scholarship. First, a foundational strand establishes that time itself can be a resource for peace: the accumulation of cooperative experience over repeated interactions helps states overcome uncertainty about adversarial intentions, building the trust and reputation necessary for stable cooperation to take hold (Osgood, 1962; Axelrod & Keohane, 1985; Hoffman, 2006; Crescenzi, 2018). We share this emphasis on the temporal dimension of interstate relations, but shift attention from the mere accumulation of cooperation to its temporal structure. Specifically, our findings suggest that it is not only the duration or quantity of past cooperative experience that determines whether optimistic expectations are formed, but also the pattern through which that experience unfolds. Moreover, we contribute to a growing body of work examining the microfoundations of public resistance to rapprochement. Existing studies find that durable peace is difficult to achieve in part because individuals bring a systematic negativity bias toward foreign adversaries to their evaluations of conciliatory gestures (Albuyeh & Paradis, 2018; Kertzer et al., 2020; S. J. Kim, 2021; Han, 2025). By showing that trajectories marked by negative changes in cooperation shape expectations of an adversary's future behavior, we identify a previously overlooked determinant of whether conciliatory efforts consolidate or unravel.

2. Temporal Dynamics of Rapprochement and the Public's Retrospective Evaluation

Rapprochement refers to the initial phase in which state leaders seek to address core disputes with a foreign adversary. It typically unfolds through high-level decisions to exchange peace gestures that build trust and pave the way for deeper cooperation, especially in managing disputes peacefully (Kupchan, 2010; Mattes & Weeks, 2024). Although rapprochement can produce rapid improvements in interstate relations by changing state leaders' behavior on a limited

set of issues (Goertz et al., 2016; Thompson et al., 2022), it is not a singular diplomatic event but an ongoing process of iterated cooperation (Kupchan, 2010; Tang, 2011; Mattes & Weeks, 2024). Figure 1 illustrates this feature: rapprochement unfolds between t_1 and t_2 , during which government-level cooperation develops through a series of cooperative exchanges between state leaders. It is precisely this temporal character that makes rapprochement a critical bridge between rivalry and conciliation — the deeper, societal-level peace extending beyond t_2 in which citizens come to embrace cooperation with a former adversary.

Figure 1: Interstate Cooperation in Rivalry, Rapprochement, and Conciliation



Historical cases further illustrate that rapprochement is defined not by a singular decisive act but by a succession of intergovernmental exchanges unfolding over time. The Nixon administration’s rapprochement with China in the early 1970s progressed through a carefully staged series of signals — the relaxation of trade restrictions, the “ping-pong diplomacy” exchanges, and ultimately Nixon’s historic visit to Beijing — each building on the last to generate momentum toward normalization (Goh, 2004). In the same way, the Egyptian-Israeli peace process was not defined by Sadat’s 1977 visit to Jerusalem alone, but by the sequence of diplomatic exchanges, confidence-building measures, and negotiating concessions that followed, culminating in the Camp David Accords (Berenji, 2020). Similarly, the gradual thaw in US-Soviet relations under Gorbachev in the late 1980s unfolded through successive rounds of arms control proposals, unilateral troop reductions, and summitry, each signal reinforcing the credibility of those that came before (Kydd, 2005). More recently, the 2018 inter-Korean rapprochement unfolded over an extended period through a succession of intergovernmental summits and cross-border cultural exchanges (Han, 2025). In each of these cases, it was the cumulative sequence of signals, not any single gesture, that constituted the process of rapprochement.

However, existing explanations of public support for rapprochement treat each cooperative

gesture as a standalone event, evaluated in isolation rather than as part of an unfolding sequence. Prior studies on reciprocated cooperation maintain that conciliatory gestures lead the public to update beliefs about rival intentions, generating greater support for conciliatory policies (Colaresi, 2004; Rasler et al., 2013; Davies & Johns, 2016). Yet they treat regime-level cooperation as a proxy for cooperation that citizens observe or perceive (Colaresi, 2004; Rasler et al., 2013) and equate a singular cooperative signal with the whole process of rapprochement (Davies & Johns, 2016). A similar limitation applies to the literature on the costliness of cooperative signals. Drawing on Kydd (2005)’s claim that bold gestures are particularly effective at signaling benign intentions under conditions of high distrust, recent empirical evidence shows that costlier signals more effectively shift individual perceptions of adversary intentions (Kertzer et al., 2020; Yoder & Haynes, 2021; S. J. Kim, 2021; Mattes & Weeks, 2022). By design, however, these studies abstract from the temporal structure of rapprochement, treating it as a singular cooperative exchange characterized by a given level of costliness and leaving unanswered how citizens evaluate cooperative signals as they accumulate over time.

In reality, citizens rarely encounter cooperative signals in isolation. Specifically, new developments during rapprochement are consistently framed by media coverage in light of prior events, prompting audiences to evaluate each signal retrospectively as part of an unfolding sequence. For instance, coverage of the Trump–Kim Hanoi summit situated its outcomes within the broader trajectory of prior meetings and conciliatory exchanges (BBC, 2018), while South Korean President Moon Jae-in’s visit to Pyongyang was widely understood in connection with the two earlier inter-Korean summits held that same year (Shin & Lee, 2018). More recently, reports on the normalization of Saudi–Iranian relations foregrounded the series of prior negotiations and gestures that made the agreement possible (Al Jazeera, 2023). In this way, media framing encourages publics to construct meaning from rapprochement as an evolving process, linking multiple cooperative signals across time to assess the broader credibility of the peace effort.

Furthermore, this retrospective character of public evaluation is theoretically grounded. Historical grievances, entrenched negative images, and threat narratives create cognitive anchors that make it difficult to reinterpret adversary motives even in the face of conciliatory acts (Jervis,

1976; Thompson, 2001; Kertzer et al., 2020), such that a single cooperative gesture rarely suffices to revise deeply held priors. Previous scholarship has long recognized this dynamic: Osgood's (1962) GRIT framework and Axelrod's (1984) iterated cooperation model both establish that trustworthiness is built through the frequency and consistency of cooperation across multiple interactions, not through any single conciliatory act. Citizens may therefore adopt a wait-and-see posture, withholding judgment until they have accumulated sufficient evidence to assess whether early cooperative signals reflect a genuine shift in the adversary's disposition or a strategically motivated departure from an otherwise hostile posture. These accumulated judgments can therefore shape whether elite-level rapprochement develops into broader public support.

We posit that such retrospective evaluation of rapprochement can shape citizens' judgments about the adversary's future behavior. At the interstate level, a large body of IR scholarship demonstrates that the trajectory of past interactions, not merely their current level, carries independent information about future behavior (Crescenzi & Enterline, 2001; Crescenzi et al., 2008; Chadeaux, 2022; Schincariol & Chadeaux, 2024). A parallel mechanism may operate at the individual level. Rather than perceiving events as isolated data points, people situate them within a temporal narrative, using sequence to infer what is likely to occur next (Cohen, 2018). The direction and internal consistency of a sequence convey information about the underlying forces driving it, creating patterns that shape anticipations of future developments (Pierson, 2004; Hom, 2020). This interpretive process — known as *emplotment* — integrates discrete events into a coherent storyline through which individuals identify continuity, causality, and purpose (Ricoeur, 1984), using the unfolding pattern of past events as the primary basis for expectations about what is likely to follow.

In the context of rapprochement, citizens use the trajectory of cooperative signals as the primary evidence from which to infer whether the adversary's cooperative behavior will endure. This inference is consequential because uncertainty about adversary intentions is structural: reading the underlying motives of a foreign adversary is inherently difficult (Schelling, 1960; Jervis, 1976; Yarhi-Milo, 2013). Rapprochement presents a particularly acute version of the challenge, as cooperative gestures may reflect a genuine and durable shift in intentions or

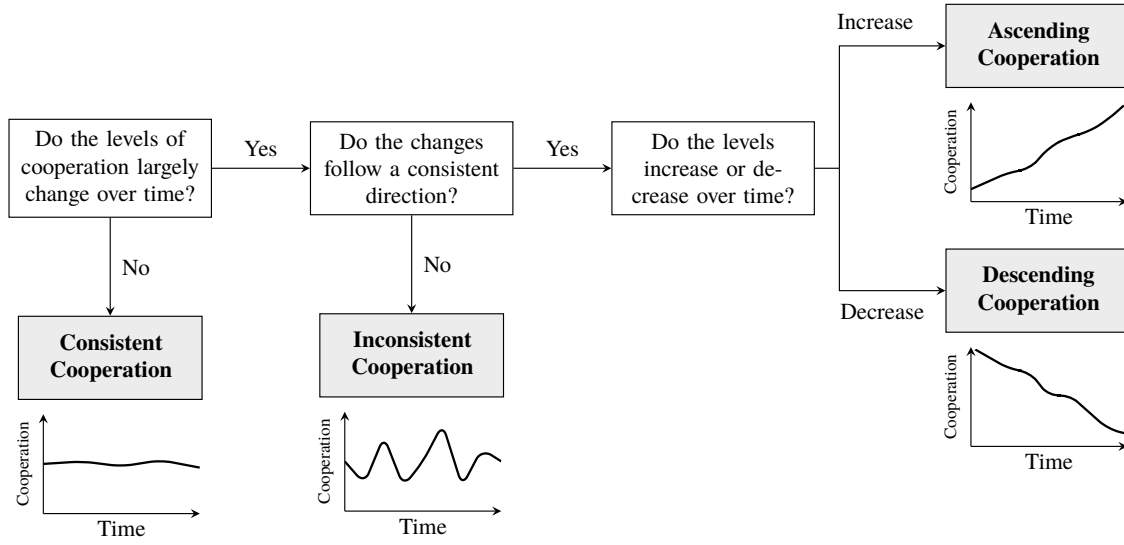
merely a strategically motivated accommodation (Kydd, 2005; Kupchan, 2010). How citizens resolve this uncertainty determines the political fate of the peace process. When the trajectory of cooperation appears consistent and enduring, citizens are more likely to view de-escalation as credible and safe; when signals appear sporadic or easily reversible, accommodation seems risky and prone to exploitation, leaving diplomatic progress confined to the elite level and vulnerable to public backlash. Identifying what types of cooperative patterns shape these expectations is therefore essential to explaining when and why peace initiatives garner sustained public support, and it is to this question that the following section turns.

3. Patterns of Cooperation and Public Opinion on Rapprochement

In this section, we identify different cooperation trajectories during rapprochement and develop testable hypotheses about their effects on public support for peace. While doing so, we focus on how the costliness of cooperative signals evolves over time. Building on signaling theory's claim that only sufficiently costly signals reliably indicate true intent (Schelling, 1960; Fearon, 1997), Kydd (2005) demonstrates that costlier acts of cooperation serve as more credible indicators of genuine cooperative intent because they are harder to fake. Applied to public perceptions of rapprochement, existing work has documented the positive role of costly signals on average (Kertzer et al., 2020; Yoder & Haynes, 2021; S. J. Kim, 2021), while subsequent studies show these effects are conditioned by individual predispositions (Kertzer et al., 2020; Fu & Lee, 2025), leaders' foreign policy reputation (Mattes & Weeks, 2022), the source of cooperative signals (Albuyeh & Paradis, 2018), and shared social identities (Han, 2025).

We propose the temporal structure of rapprochement as an important yet underexplored factor that conditions the effects of costly signals. Citizens may evaluate rapprochement not only by the costliness of individual signals but also by how costliness changes over time. Specifically, each signal is not evaluated in isolation — prior signals serve as reference points against which subsequent ones are assessed. Even when the overall level of cooperation remains positive, a shift toward less costly signals can register as a relative decline, shaping citizens' perceptions

Figure 2: Types of Cooperation Trajectories during Rapprochement



of the trajectory independently of its aggregate value.

We argue that such declines carry particular evaluative weight due to the prior belief structure that citizens bring to rivalry contexts. Long-standing enmity leaves citizens with entrenched expectations that the adversary is untrustworthy and that hostility is the default state of the relationship (Jervis, 1976; Thompson, 2001), predisposing them to interpret incoming signals through existing hostile priors rather than revise them (Kertzer et al., 2020; Han, 2025). As a result, even when a rapprochement record shows net positive cooperation, a decline in costliness between signals can reactivate citizens’ underlying image of the adversary as fundamentally uncooperative. That is, because such declines are consistent with preexisting expectations of adversarial behavior, they carry disproportionate weight relative to equivalent gains — suppressing expectations of future cooperation and undermining perceptions of successful rapprochement.

This theoretical logic leads us to identify specific cooperation patterns that dampen the effects of the olive branch. As shown in Figure 2, we organize these trajectories into a typology along two dimensions: (1) whether the level of cooperative signaling changes significantly over time and, (2) if so, the direction of that change.

The first distinction separates trajectories in which the level of cooperative signals remains relatively constant from those in which it varies. When cooperation remains stable, the trajectory is classified as *consistent cooperation*; when it varies without a clear directional pattern,

as *inconsistent cooperation*. These two patterns are illustrated by the cases of Franco–German reconciliation after World War II and Indo-Pakistani relations in the 2000s. In postwar Europe, France and Germany engaged in a steady, institutionalized process of collaboration—from the European Coal and Steel Community to the broader European Union framework—that gradually built trust and made peace self-reinforcing (Germond & Türk, 2008). Indo-Pakistani relations offer a contrasting illustration: the Composite Dialogue process, initiated in 2004, produced meaningful progress across trust-building measures, trade, and territorial disputes, but remained highly vulnerable to disruption, leaving cooperation oscillating between moments of engagement and prolonged stalemate (Wojczewski, 2014).

Citizens may observe similar aggregate levels of cooperation across these two trajectories, yet inconsistent cooperation exposes them to recurring episodes of decline. Research on retrospective evaluation suggests that assessments of extended sequences are disproportionately shaped by salient moments — especially peak and end experiences — rather than by the integration of all available information (Kahneman et al., 1993; Redelmeier & Kahneman, 1996; Ariely & Carmon, 2000). Relatedly, work on belief updating shows that negative feedback can disrupt updating processes even when positive information continues to accumulate (Zimmermann, 2020). Similarly, declines in cooperation are particularly likely to resonate with preexisting expectations of adversarial behavior. By confirming hostile priors, such episodes acquire a disproportionate evaluative weight relative to equivalent gains in cooperation. Consequently, citizens exposed to an inconsistent trajectory are likely to place greater weight on episodes of deterioration than on periods of stability, leading them to form less optimistic expectations about the adversary’s future behavior. We therefore hypothesize that, relative to a consistent trajectory, an inconsistent trajectory will reduce expectations of future cooperation (*H1A*) and decrease support for conciliatory policies toward the adversary (*H1B*).

***H1A:** Exposure to an inconsistent trajectory of cooperation, compared to a consistent trajectory, leads individuals to expect lower future cooperation from a foreign adversary.*

***H1B:** Exposure to an inconsistent trajectory of cooperation, compared to a consistent trajectory, decreases individuals’ support for a conciliatory policy toward a foreign adversary.*

When changes in the level of costly cooperation do follow a systematic direction, the trajectory is categorized according to whether cooperation becomes stronger or weaker over time. *Ascending cooperation* describes a pattern in which the level of cooperative signals gradually increases, whereas *descending cooperation* refers to a pattern in which cooperation progressively declines. Historical cases show the difference between these two types of cooperation patterns. Following normalization in 1995, US-Vietnam relations exhibited an ascending cooperation trajectory, with cooperation gradually expanding from diplomatic engagement to broader economic and security ties (Siracusa & Nguyen, 2017). By contrast, recent inter-Korean relations offer an example of descending cooperation, as the momentum generated by the 2018 summits and cooperative initiatives gradually dissipated amid stalled negotiations and limited progress on denuclearization (Kwon, 2022).

We expect a descending trajectory to dampen public support for peace relative to an ascending one, because a series of reductions in cooperation accumulate over time. Research on retrospective evaluation shows that judgments of extended sequences depend not only on the overall level of outcomes but also on their direction of change: sequences that improve over time are evaluated more favorably, whereas sequences that deteriorate over time are evaluated more negatively (Hsee & Abelson, 1991; Karney & Frye, 2002).

In the context of rapprochement, descending trajectories repeatedly reactivate hostile priors by exposing individuals to a pattern of deteriorating cooperation, whereas ascending trajectories gradually challenge adversarial expectations through a pattern of successive improvements. We therefore hypothesize that, compared with exposure to an ascending trajectory, exposure to a descending trajectory will lead individuals to form lower expectations of future cooperation (H2A) and to express lower support for conciliatory policies toward the adversary (H2B).

***H2A:** Exposure to a descending trajectory of cooperation, compared to an ascending trajectory, leads individuals to expect lower future cooperation from a foreign adversary.*

***H2B:** Exposure to a descending trajectory of cooperation, compared to an ascending trajectory, decreases individuals' support for a conciliatory policy toward a foreign adversary.*

4. Research Design Overview

To test our argument, we conducted survey experiments in the U.S. and South Korea. This methodological choice offers several advantages particularly well-suited to studying how patterns of cooperation across time shape public opinion on rapprochement. First, observational data rarely offer the variation needed to test how cooperation trajectories shape public opinion. Cross-national cases of rapprochement are too few to permit systematic comparison, and public opinion data aligned with distinct trajectory patterns are difficult to obtain. Survey experiments address both limitations by allowing us to construct the variation in cooperation trajectories and to observe how publics respond to them differently. Second, because the timing of rapprochement is often strategically chosen, observational designs face serious selection problems (Mattes & Weeks, 2019, 2022). That is, cooperation trajectories are themselves likely endogenous to prior political conditions, making experimental leverage essential for causal identification. Lastly, survey experiments allow us to isolate the effect of cooperation trajectories from confounding individual-level characteristics—such as prior attitudes toward the adversary, political ideology, or partisan identity—that would otherwise obscure the mechanisms driving opinion change (Kertzer & Tingley, 2018).

Our first survey experiment (Study I) examines how Americans perceive China and the prospect of bilateral cooperation. China’s rise has intensified geopolitical frictions with the U.S. (Kertzer et al., 2024), fueling deep-seated public hostility and negative stereotypes that pose significant domestic barriers to reconciliation (Xu, 2025). Given the salience of this rivalry and the risks it poses to global stability, a growing body of research has examined public opinion on foreign policy amid rising U.S.-China tensions (Mattes & Weeks, 2019; Myrick, 2021; Kertzer et al., 2024; Han et al., 2026). This experiment contributes to this strand of literature by addressing how different cooperation trajectories shape Americans’ support for rapprochement with China. Specifically, we presented respondents with a hypothetical scenario depicting potential future U.S.-China tensions, with variation in the patterns of China’s cooperative signals embedded in the scenario. Because the design is forward-looking and abstract, it separates the treatment effects from pre-existing attitudes, partisan narratives, or memories of specific events—which

lets us isolate the theoretical mechanisms from other confounding factors (Brutger et al., 2023). As such, this method has become a common approach in recent IR literature that investigates the microfoundations of cooperation between foreign adversaries (Davies & Johns, 2016; Mattes & Weeks, 2019, 2022; S. J. Kim, 2021; Lee, 2023; Kertzer et al., 2024).

However, this experiment has several limitations worth acknowledging. First, American respondents have little lived experience with great-power rapprochement, raising concerns about ecological validity. That is, their reactions may reflect unfamiliarity with the scenario rather than genuine updating on cooperative signals. Second, while prior work suggests that hypothetical framing leaves treatment effects largely unchanged (Brutger et al., 2023), our design presents cooperative signals in a sparse information environment—stripped of the competing political considerations that characterize real-world foreign policy contexts—which may cause us to overestimate the effect of cooperation trajectories. Lastly, our hypothetical scenario measures support for conciliation in the abstract rather than for the specific, politically contentious policies that define the current bilateral agenda, potentially overstating public willingness to cooperate in practice.

To address these limitations, we turn to our second survey experiment in South Korea (Study II), where rapprochement is not an abstract or hypothetical scenario but a lived and ongoing political reality. Inter-Korean relations offer an ideal setting for this purpose: the peninsula has seen recurring cycles of tension and engagement (Erästö & Su, 2026). Additionally, inter-Korean rapprochement remains an active and contested issue in South Korean public discourse (E. Kim, 2026), with debates over engagement policy toward the North continuing to shape domestic political alignments and electoral competition to the present day. Thus, while the U.S.–China experiment tests our theory in a context of great-power rivalry with little precedent for rapprochement, the experiment in South Korea grounds our argument in a case where respondents have direct familiarity with episodes of cooperation between rivals.

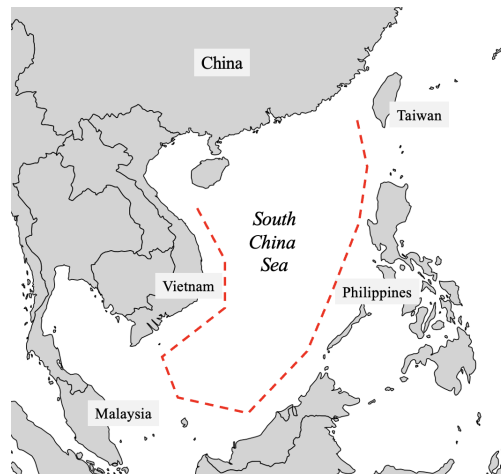
To leverage this historical episode experimentally, we employ a selective-history design in which respondents are presented with alternative narrative descriptions of North Korea’s behavior during the 2018 rapprochement. That is, we depict North Korea’s cooperative signals toward South Korea as varying over time. In addition, rather than presenting cooperation trajectories

in a sparse, abstract setting, this design manipulates the interpretive frame of a shared historical memory (Tingley, 2017; Weiss & Dafoe, 2019; Dafoe et al., 2022). Respondents evaluate cooperation trajectories rooted in events they actually experienced, embedded in the richer political context of an ongoing and contested national debate. This approach substantially enhances ecological validity, though we acknowledge that prior knowledge of those events may attenuate treatment effects compared to a hypothetical design.

5. Study I: Survey Experiment in the U.S.

Research Design

Figure 3: A Hypothetical Scenario of U.S.-China Tension



The U.S. faces major security challenges due to its tense relationship with China, particularly in the South China Sea. The region is strategically vital. Not only does one-third of global maritime trade pass through it, but it also contains valuable natural resources essential for energy security and economic stability.

China asserts its historical claims to the region and maintains artillery facilities there. The U.S. Navy regularly carries out operations by positioning military ships near disputed waters. The presence of their military forces in the South China Sea remains a major flash point, increasing the risk of escalation between the two countries.

In this study, we carried out a preregistered survey experiment on 2,324 American adults via Cint Theorem’s platform between July and September 2025.¹ At the beginning, respon-

¹The experimental design received ethics approval from the authors’ home institution, and it is pre-

dents were asked a series of demographic questions, including gender, age, religion, household income, education level, partisan identification, political ideology, and a feeling thermometer toward China.² Following the demographic and attitudinal measures, respondents were presented with a hypothetical scenario involving a potential future situation between the U.S. and China.³ As shown in Figure 3, this scenario is conveyed through a vignette accompanied by a visual treatment.

We presented respondents with a scenario in which U.S. and Chinese leaders take steps to ease regional tensions through cooperation. Participants were randomly assigned to one of six groups. The first group serves as a control condition, receiving information about the South China Sea situation in the absence of any cooperative behavior from China. This condition is essential for establishing a baseline, allowing us to identify whether exposure to any pattern of cooperation—regardless of its consistency or direction—shifts public attitudes relative to a status quo of no engagement. The control vignette reads as follows:

[Control] No Cooperation

Suppose that, in the year 2028, there is no meaningful shift in the situation. Beijing and Washington do not hold any formal talks regarding their military presence in the South China Sea, and China keeps its number of artillery facilities steady at 120 in that year.

The second group (*Treatment 1*) serves as a baseline condition of cooperation, in which participants are informed that China engaged in cooperative behavior following high-level talks with the U.S. in 2028, but without any indication of a specific pattern or trajectory. Participants learn that between May and December 2028, China dismantled a significant number of artillery facilities in the South China Sea, reducing the total from 120 to 57—but receive no information about how this reduction unfolded over time. This condition is theoretically important because it allows us to isolate the added effect of trajectory information beyond cooperation per se. That is, by comparing this group to the patterned treatment conditions, we can assess whether it is

registered at Open Science Framework (anonymous link: https://osf.io/2xw5c/overview?view_only=83e92470a2a342adad8d0f349b80d538).

²The sample's demographic characteristics are nationally representative, and our analyses focus on the respondents who passed pre-treatment attention checks. See Appendix 1.1 and 1.4 for sample demographics and the full survey questionnaires.

³We chose the year 2028 for the scenario to ensure it feels proximate and realistic, while avoiding potential bias from the current U.S. political context.

the mere fact of cooperation or the specific way it unfolds over time that drives public opinion. The vignette reads as follows:

[Treatment 1] Cooperation without a Pattern

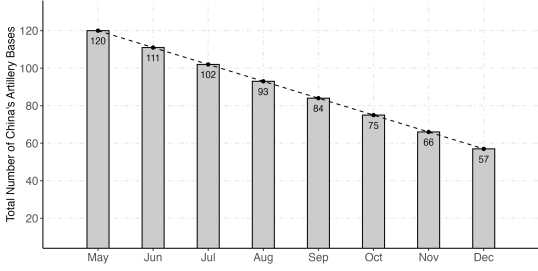
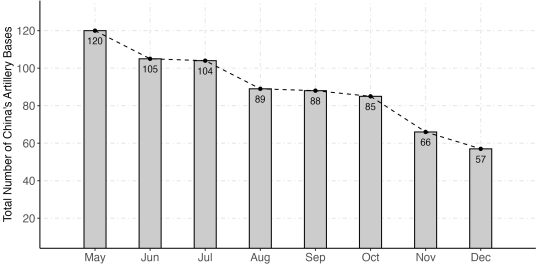
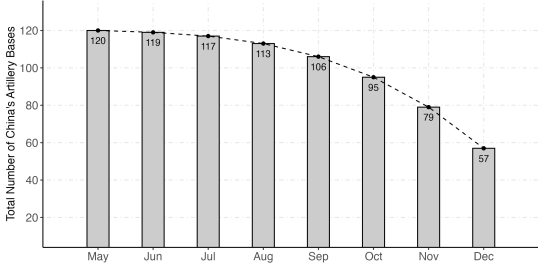
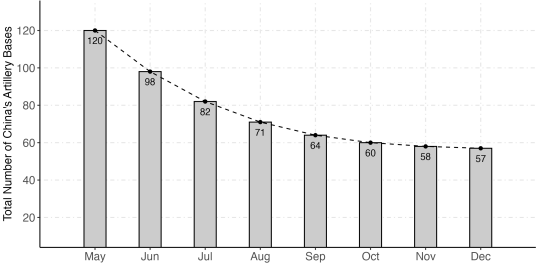
Suppose that, in the year 2028, Beijing and Washington hold high-level talks to reduce tensions in the region. Following these talks, China begins to take down its artillery facilities in the South China Sea. Between May and December in that year, the total number drops by over half, from 120 to 57.

Each of the remaining four treatment groups presents a distinct pattern of cooperative behavior from China, built upon the same basic cooperative scenario as *Treatment 1* but augmented with both textual and visual information depicting how cooperation unfolded over time.⁴ As summarized in [Table 1](#), these four conditions are designed to capture two key dimensions of cooperation trajectories: consistency—whether cooperative behavior is sustained at a stable level or fluctuates irregularly—and direction—whether cooperation is increasing or decreasing over time. Importantly, all four treatments present the same aggregate outcome: China dismantled a total of 63 artillery facilities in the South China Sea over an eight-month period following high-level talks with the U.S. in 2028, reducing the total from 120 to 57. Holding this net outcome constant across conditions ensures that any differences in public response can be attributed to the trajectory of cooperation rather than its overall magnitude.

The treatments vary only in the temporal pattern of this reduction. *Treatment 2* depicts a consistent pattern, with China removing exactly nine facilities each month, while *Treatment 3* presents an inconsistent pattern, in which the number of dismantled facilities fluctuates widely from month to month. *Treatment 4* describes an ascending trend, with few facilities removed in the early months and a sharp acceleration in dismantling activity toward the end, while *Treatment 5* presents a descending pattern, showing a strong initial reduction that tapers off over time. Together, these four conditions allow us to examine how different cooperation trajectories—despite producing identical net outcomes—shape public perceptions of China’s intentions and support for continued engagement.

⁴Following [Healy & Lenz \(2014\)](#), who use bar plots depicting yearly income growth to show how temporal patterns shape retrospective evaluations, both experimental studies in this article presented survey respondents with similar visualizations of how cooperation unfolded over time.

Table 1: Treatment Conditions with Varying Patterns of Cooperation in Study I

[Treatment 2] Consistent Cooperation  <i>China shows a steady rate of facility reduction. Each month, it consistently dismantles nine facilities from start to finish.</i>	[Treatment 3] Inconsistent Cooperation  <i>China shows an inconsistent pace of facility reduction. In some months, it dismantles more than a dozen facilities, while in others, it removes only a few.</i>
[Treatment 4] Ascending Cooperation  <i>China shows an increasing pace of facility reduction. It dismantles only a few facilities in the early months but significantly increases the number removed in the later months.</i>	[Treatment 5] Descending Cooperation  <i>China shows a declining pace of facility reduction. While it dismantles more than a dozen facilities in the early months, only a few facilities are removed in the later months.</i>

To assess the impact of the experimental treatments on participants' expectations and policy preferences, we included a series of outcome measures related to China's future behavior and U.S. responses. First, participants were asked to anticipate China's actions in the year following the scenario, specifically whether China would increase, decrease, or maintain the number of artillery facilities in the South China Sea. Those who predicted a change were then asked to estimate the magnitude of that change by indicating the number of facilities they expect China to add or remove. To assess whether their responses reflected an underlying reading of the cooperative trend, a follow-up open-ended question invited participants to explain the rationale behind their prediction. Furthermore, participants were presented with a policy question regarding the appropriate U.S. response. They were told that the U.S. currently deploys 80 military ships to

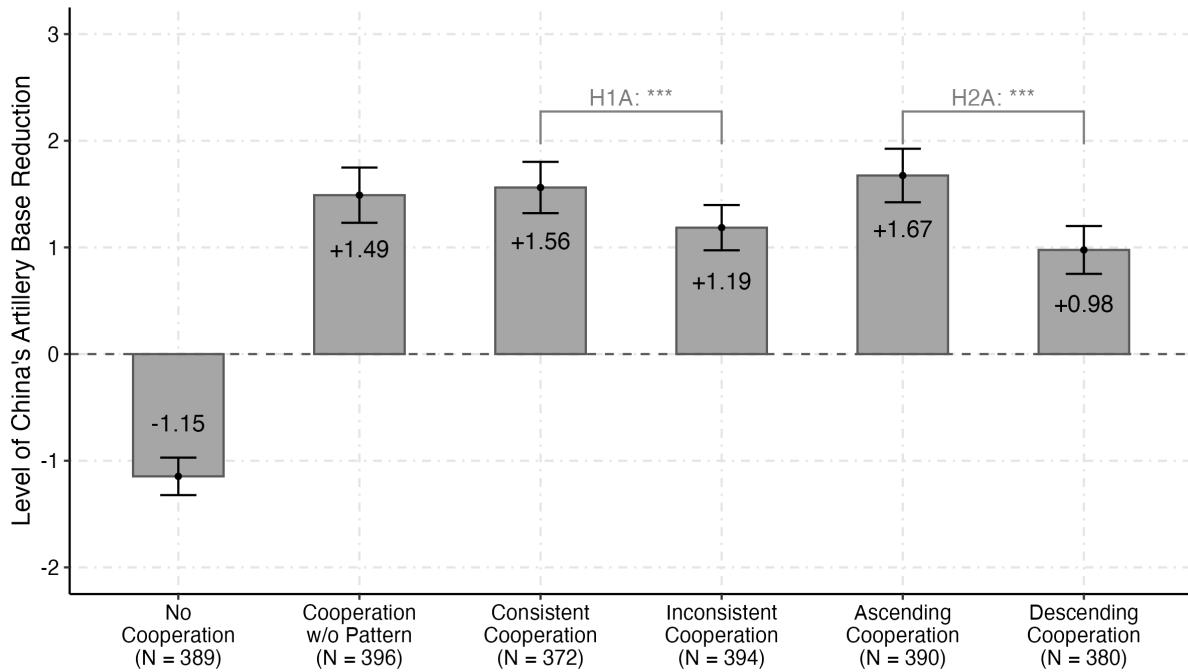
the South China Sea and were asked to indicate whether they believed this number should increase, decrease, or stay the same in light of China's actions. Those who proposed a change were then asked to estimate the magnitude of that change by indicating the number of the U.S. ships they expect its leaders to add or remove.

Results

Our initial analysis examines American respondents' expectations regarding China's future behavior in the South China Sea, as displayed in [Figure 4](#). The outcome variable is a net reduction score ranging from -5 (expectation of an expansion by 21 or more facilities) to 5 (expectation of a reduction by 21 or more facilities), such that a one-unit increase corresponds to approximately five additional facilities expected to be withdrawn. The first coefficient bar shows that respondents assigned to the no-cooperation condition (in which China removes no artillery bases) report a mean score of -1.15 , indicating that they expect China to increase its military presence in the region by approximately 5.75 facilities. By contrast, the second point estimate shows that respondents who observe a cooperative signal without any temporal pattern expect a reduction of approximately 7.45 facilities (1.49 points), suggesting that the presence of a cooperative signal from an adversary leads publics to anticipate more conciliatory behavior in the future.

Our theory predicts, however, that the trajectory of cooperation across time shapes these expectations beyond the mere presence of cooperative signals. Consistent with [H1A](#), which holds that inconsistent cooperation should generate less optimistic expectations of future adversary behavior than consistent cooperation, the third and fourth columns of [Figure 4](#) support this prediction. Respondents who observed a consistent cooperative pattern report a mean score of 1.56, while those exposed to an inconsistent pattern report 1.19 points — a gap of roughly 1.85 facilities that is statistically significant at the 0.01 level. This difference is largely driven by the dampening effect of inconsistent cooperation rather than a boost from consistent cooperation: consistent cooperation raises expectations by only 0.07 relative to the patternless cooperation, falling short of conventional significance ($p = 0.69$), whereas inconsistent cooperation reduces expectations by 0.3 points ($p = 0.07$).

Figure 4: Patterns of Cooperation and Expectation of China's Cooperation

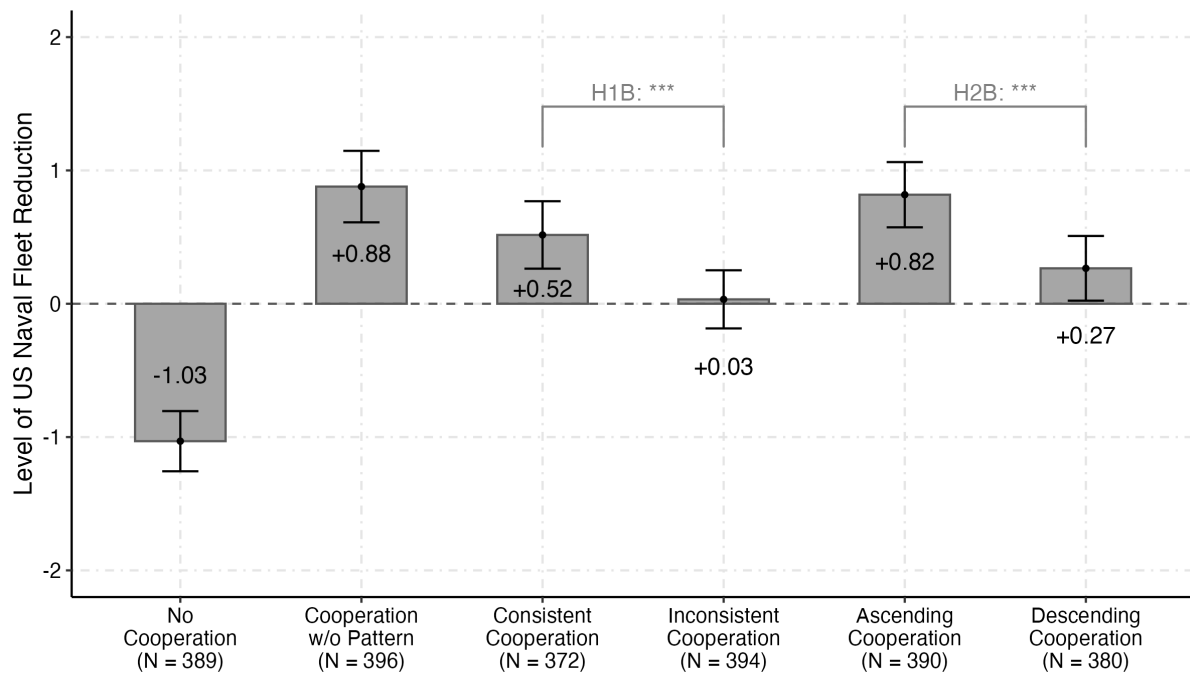


Note: Error bars represent 95 percent confidence intervals. ***: $p < 0.01$; **: $p < 0.05$; *: $p < 0.1$.

A similar pattern emerges along the directional dimension, though with larger effects. Comparing the fifth and sixth columns in the figure provides support for H2A, which predicts that descending cooperation produces less optimistic expectations of future adversary behavior than ascending cooperation. Although respondents in both conditions anticipate a net reduction in China's military facilities, respondents exposed to an ascending cooperative pattern report a mean score of 1.67, compared to 0.98 among those exposed to the descending condition — a difference of approximately 3.45 facilities that is statistically significant at the 0.01 level. As with the consistency dimension, decomposing this difference reveals that it is driven primarily by the negative effect of descending cooperation: while ascending cooperation moderately increases cooperative expectations by 0.18 points relative to the patternless baseline ($p = 0.31$), descending cooperation significantly reduces expectations by 0.51 points ($p < 0.01$).

Turning to policy attitudes, Figure 5 displays American respondents' opinions on the appropriate U.S. military response, measured as the preferred shift in the number of U.S. naval vessels deployed in the South China Sea relative to a baseline of 80 ships. The outcome variable is a net withdrawal score ranging from -5 (an increase of 21 or more vessels) to 5 (a reduction of

Figure 5: Patterns of Cooperation and Public Opinion on the U.S. Conciliatory Policy



Note: Error bars represent 95 percent confidence intervals. ***: $p < 0.01$; **: $p < 0.05$; *: $p < 0.1$.

21 or more vessels), with each one-unit increase corresponding to approximately five additional ships withdrawn. Respondents in the condition with no cooperative signals from China report a mean score of -1.03 , indicating a preference to increase the presence of the US Navy in the region by roughly 5 ships. This preference reverses when a cooperative signal is introduced without a discernible temporal pattern. Respondents in the patternless cooperation condition favor a reduction of approximately 4 ships ($+0.88$ points), indicating that the observation of adversarial cooperation shifts public demand toward a more conciliatory policy posture.

The trajectory of cooperation again shapes policy preferences beyond the mere presence of cooperative signals. The third and fourth columns of Figure 5 support H1B, which predicts that inconsistent cooperation generates lower public support for conciliatory policy than consistent cooperation. Respondents exposed to the consistent trajectory report a mean score of 0.52 , compared to 0.03 among those in the inconsistent condition — a difference that is statistically significant at the 0.01 level. Notably, the inconsistent condition's mean is indistinguishable from zero in this measure (0.03), suggesting that when cooperative signals are erratic, the public remains unwilling to respond with reciprocal concessions despite having observed some degree

of adversarial cooperation. This gap is driven more by the suppressive effect of inconsistency than by a positive boost from consistent cooperation. While inconsistent cooperation reduces support for conciliatory policy by 0.85 points relative to the patternless baseline ($p < 0.01$), consistent cooperation also reduces support by 0.36 points relative to the patternless baseline ($p < 0.1$). This is a pattern that, while counterintuitive, may reflect the possibility that a mechanically regular pattern of cooperation paradoxically invites suspicion rather than reassurance, as respondents may interpret the perfectly steady cadence of cooperative acts as evidence of deliberate strategic calculation rather than genuine conciliatory intent.

A larger effect of cooperation trajectories emerges along the directional dimension. The fifth and sixth columns in this figure support [H2B](#), which predicts that descending cooperation should produce lower public demand for conciliatory policy than ascending cooperation. Respondents exposed to an ascending pattern report 0.82 points on average, compared to only 0.27 among those in the descending condition — a difference of approximately 2.75 vessels that is statistically significant at the 0.01 level. This gap is again driven primarily by the suppressive effect of descending cooperation: ascending cooperation shifts preferences by a negligible margin relative to the patternless baseline (0.06 points, $p = 0.74$), whereas descending cooperation reduces support for withdrawal by 0.61 points, an effect that is statistically significant ($p < 0.01$).

We conducted several robustness checks. First, to verify that respondents engaged with the cooperation patterns presented in treatments, we analyzed open-ended responses asking them to explain their expectations of China’s future behavior, coding each as 1 if it referenced temporal cooperation patterns and 0 otherwise. Of 2,321 respondents, 855 drew explicitly on pattern-based reasoning. As shown in Appendix 1.2, those assigned to any of the four patterned treatment conditions were significantly more likely to invoke temporal reasoning than those in the patternless or no-cooperation conditions. This suggests that our treatment effects reflect respondents’ active interpretation of cooperation trajectories rather than the mere presence of cooperative signals. Additionally, as demonstrated in Appendix 1.3, a balance check confirms that the experimental groups do not differ significantly from one another in terms of participants’ demographic characteristics.

6. Study II: Survey Experiment in South Korea

Research Design

In our second study, we conducted a preregistered survey experiment in South Korea.⁵ Between January and April 2026, we recruited 2,000 participants through Macromill Embrain (<https://www.embrain.com/eng/>), a South Korean online survey panel provider.⁶ The main goal of this study is to investigate how exposure to priming information on different patterns of cooperation during the 2018 inter-Korean rapprochement shapes South Korean respondents' expectations of North Korea's future behavior and their support for conciliatory policies. While our primary interest lies in comparing the effects of opposing trajectory patterns along the dimensions of consistency and direction, we also examine the extent to which exposure to these patterns shifts respondents' perceptions of North Korea relative to their pre-existing baseline. To facilitate this, we included a pure control group that received no information about the relevant event, and designed a second experimental group (*Treatment 1*) in which respondents were exposed to information about the 2018 inter-Korean rapprochement without any emphasis on a specific cooperative pattern. For the treatment's priming, the corresponding vignette enumerates the major diplomatic, cultural, and military exchanges of that year without highlighting any temporal trajectory. This allows us to test whether it is exposure to cooperation patterns specifically—rather than mere awareness of the rapprochement episode—that drives respondents' evaluations in a particular direction. The vignette reads as follows:

[Treatment 1] Cooperation without Pattern

Throughout 2018, North Korea pursued various forms of cooperation toward South Korea. In his New Year's address that year, Chairman Kim Jong-un mentioned the possibility of participating in the PyeongChang Winter Olympics, leading to the reopening of inter-Korean communication channels that had long been suspended. Subsequently, North Korea carried out a range of cooperative measures, including exchanges of high-level delegations, consultations on family reunions, discussions on easing tensions along the Military Demar-

⁵The study received ethics approval from the authors' home institution and was preregistered on the Open Science Framework (anonymous link: https://osf.io/adnve/overview?view_only=208de0e4d77a4088a8139f13bb6bbf96).

⁶Only respondents who (1) agreed to all three consent statements (participation, sensitive data use, and debriefing), and (2) correctly answered pre-treatment attention check questions are included in the final dataset. See Appendix 2.6 for the full questionnaires.

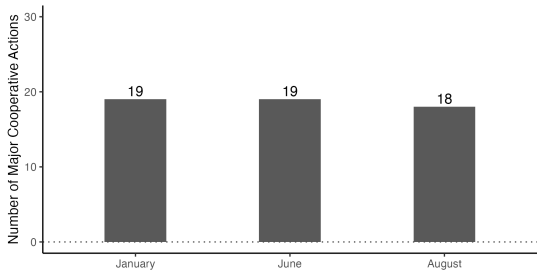
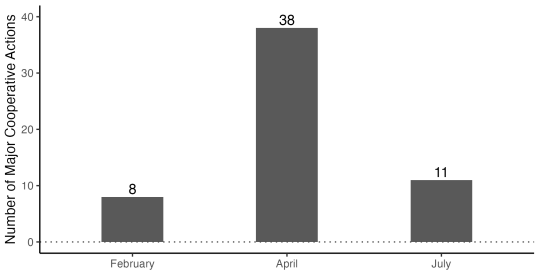
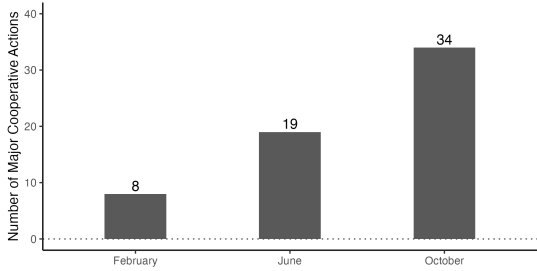
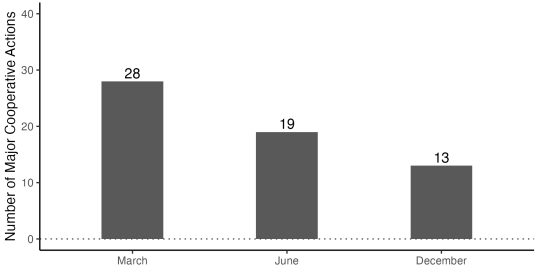
cation Line, the signing of a military agreement, the formation of a joint inspection team for railway connections, and verification of DMZ demilitarization.

As summarized in [Table 2](#), each treatment group with a distinct cooperative pattern was presented with a vignette and accompanying image describing inter-Korean relations in 2018.⁷ *Treatment 2* and *Treatment 3* show contrasting narratives along the dimension of cooperation's consistency. The treatment of consistent cooperation depicts 2018 as a sustained period of engagement in which North Korea repeatedly followed through on diplomatic, cultural, and military confidence-building initiatives, portraying cooperation as stable across time. The treatment of inconsistent cooperation, by contrast, frames the same year as marked by irregular engagement: while acknowledging key cooperative gestures, it highlights delays, pauses, and uneven follow-through, presenting North Korea's behavior as intermittent rather than sustained. On the other hand, *Treatment 4* and *Treatment 5* present opposing temporal narratives along the dimension of cooperation's direction. The treatment of ascending cooperation frames the inter-Korean engagement as an intensifying trajectory, beginning with symbolic gestures and evolving into increasingly institutionalized cooperation, culminating in the September Pyongyang Summit agreements—emphasizing an upward arc in which cooperation broadened in scope and frequency over time. The treatment of descending cooperation, by contrast, frames 2018 as beginning with dramatic overtures that gradually lost momentum, shifting from symbolic breakthroughs and high-profile summits to more routine implementation of earlier agreements, thereby highlighting a downward slope in the intensity of North Korea's engagement.

Additionally, each treatment vignette combines the verbal narrative with quantitative data and a visual trend drawn from the Integrated Crisis Early Warning System (ICEWS), a repository of event-data maintained by the U.S. Department of Defense that systematically records cooperative and conflictual actions between states ([Boschee et al., 2015](#)). We incorporate this visualization because verbal narratives alone may be insufficient to convey the temporal structure of cooperation trajectories. The ICEWS data and accompanying graph anchor each narrative in a documented empirical record, enhancing treatment uptake and making the trajectory easier for respondents to interpret.

⁷See Appendix 2.1 for the full version of the treatment vignettes.

Table 2: Treatment Conditions of Cooperation with Different Patterns in Study II

[Treatment 2] Consistent Cooperation  Throughout 2018, North Korea displayed a consistent pattern of cooperation toward South Korea. Early in the year, the two Koreas exchanged high-level delegations ... This cooperation subsequently extended to discussions in July on easing tensions around the Military Demarcation Line ...	[Treatment 3] Inconsistent Cooperation  In 2018, North Korea exhibited an inconsistent pattern of cooperation toward South Korea ... the intensity of cooperation declined ... and in October cooperation was limited to more constrained initiatives ... In November, cooperation increased again ...
[Treatment 4] Ascending Cooperation  In 2018, North Korea's cooperation toward South Korea gradually intensified over time. Early in the year, cooperation remained largely at the level of signaling willingness to engage. ... Cooperation deepened further with the conclusion of a military agreement in September ...	[Treatment 5] Descending Cooperation  In 2018, North Korea's cooperation toward South Korea gradually weakened over time. Early in the year, a relatively high level of cooperation was evident ... Toward the end of the year, cooperation became largely confined to maintaining communication channels through the inter-Korean liaison office ...

Following the treatment stage, respondents entered a measurement stage in which they answered two sets of questions capturing our key outcome variables, each measured on five-point Likert scales. The first set measures expectations about North Korea's future behavior, asking respondents to evaluate two dimensions beyond 2026: (1) the military posture they expect North Korea to adopt on the Korean Peninsula, and (2) the stance they anticipate North Korea will take toward diplomatic exchanges with South Korea. The second set measures attitudes toward South

Korea's policy choices, asking respondents to indicate (1) the level of military deployment and surveillance they believe South Korea should maintain along the inter-Korean border, and (2) their support for expanding exchanges and cooperation with North Korea.

After completing the outcome measures, respondents were asked to select the description that best captured North Korea's cooperative pattern in 2018 ("consistent," "inconsistent," "increasing," "decreasing," or "don't know"). This item functioned as a manipulation check, allowing us to verify whether participants correctly understood and internalized the temporal cooperation pattern emphasized in their assigned treatment. We also included a question asking respondents to characterize North Korea's policies toward South Korea from 2019 to 2026, using a scale from "very cooperative" to "very hostile." This item assesses whether exposure to differently framed cooperation trajectories shapes not only forward-looking expectations but also retrospective interpretations of North Korea's subsequent behavior. Such backward-looking evaluations matter because perceptions of the 2018 rapprochement, the post-2018 period, and future expectations may form a connected evaluative chain, with the assigned trajectory coloring how respondents read the entire arc of inter-Korean relations.

At the conclusion of the survey, respondents reported a set of background characteristics, including political ideology, partisan identification, household income, residential location, and educational level. All respondents then received a debriefing statement explaining the purpose of the study. This stage was designed to clarify that the survey vignettes were based on actual inter-Korean events from 2018 but were intentionally framed in different ways to examine how temporal narratives of cooperation shape public perceptions. Only respondents who indicated that they understood the debriefing's explanation and consented to the use of their responses were retained in the final sample.

Results

We first analyze how South Korean respondents expect North Korea to behave in the future, as displayed in [Figure 6](#). We use two outcome variables measuring expected cooperation, measured on a 5-point Likert scale (0–4): (1) the level of North Korea's military disengagement on

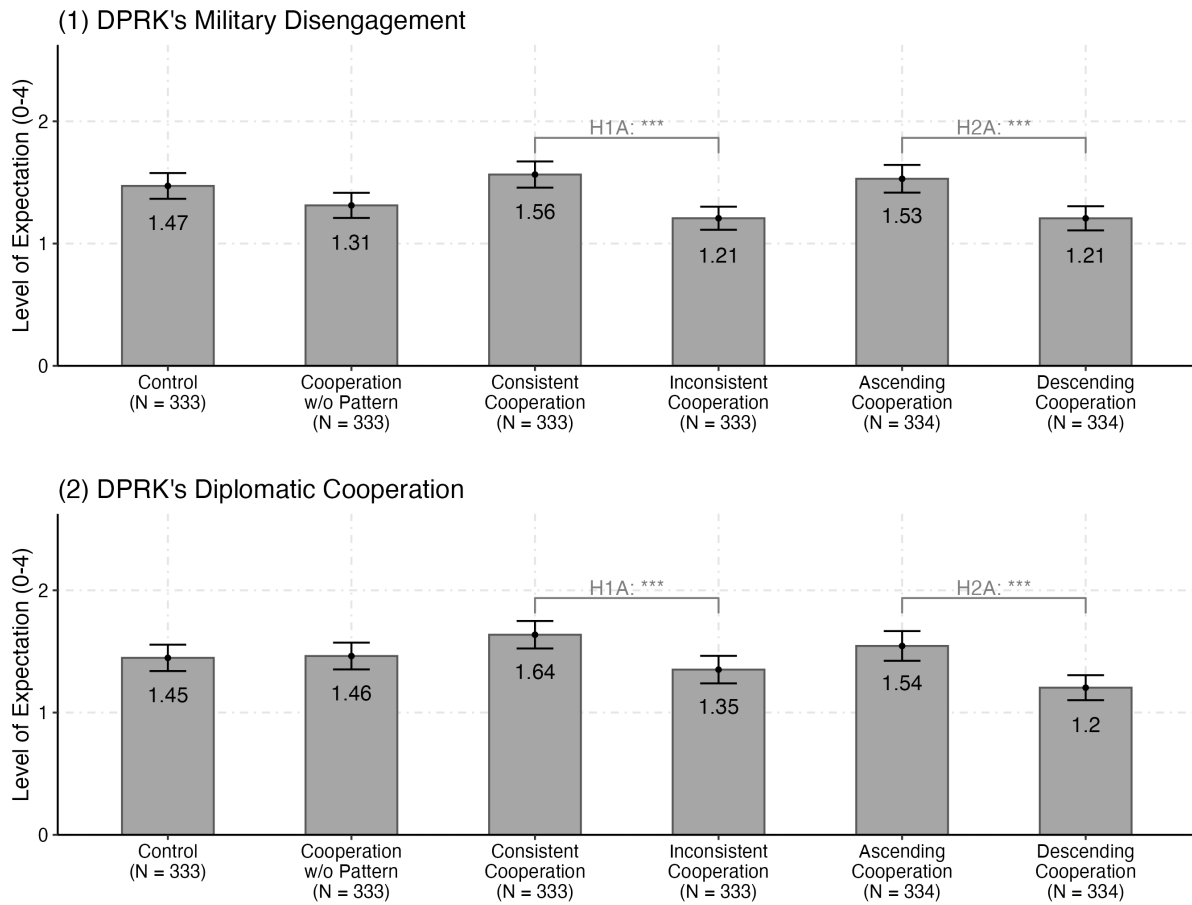
the Korean Peninsula, and (2) the level of North Korea's diplomatic cooperation toward South Korea. As seen in the first coefficient bar of the two graphs, respondents assigned to the pure control condition, in which no information regarding North Korea's past cooperation was provided, report mean scores of 1.47 and 1.45, respectively, suggesting that respondents on average anticipated continued tension rather than cooperation.

Notably, in both graphs, the second point estimate shows that respondents exposed to a description of North Korea's 2018 cooperation without any temporal pattern did not exhibit a significantly higher expectation of concessions. In fact, for the first outcome variable regarding North Korea's military disengagement, they are less likely to expect North Korea to decrease military tensions on the Korean Peninsula with a mean difference of 0.16 ($p < 0.05$). This stands in contrast to the first study's hypothetical setting, where exposure to cooperative behavior alone was sufficient to shift expectations in a positive direction. With a real historical adversary, reminders of past cooperation may also activate deeply entrenched negative perceptions shaped by decades of conflict. Relative to the no-information control, making North Korea's behavior salient may therefore reinforce hostile priors and attenuate—or even reverse—the positive effect of cooperative signals.

Our theory predicts that the trajectory of cooperation across time shapes expectations beyond the mere presence of cooperative signals. Congruent with [H1A](#), which holds that inconsistent cooperation should generate less optimistic expectations of future adversary behavior than consistent cooperation, the third and fourth columns in this figure support this theoretical expectation. For both outcome variables, respondents exposed to a consistent cooperative pattern report higher mean scores than those in the inconsistent condition — 1.56 versus 1.21 for military disengagement and 1.64 versus 1.35 for diplomatic cooperation — with both differences statistically significant at the 0.01 level.

Across both outcomes, the direction of the estimates is consistent: consistent cooperation increases expectations relative to the control group, whereas inconsistent cooperation decreases them. For military disengagement, consistent cooperation produces a positive but statistically insignificant increase, with a mean difference of 0.09 ($p = 0.22$), while inconsistent cooperation significantly reduces expectations, with a mean difference of -0.26 ($p < 0.01$). For diplomatic

Figure 6: Patterns of Cooperation and Expectations of North Korea's Future Military and Diplomatic Cooperation.



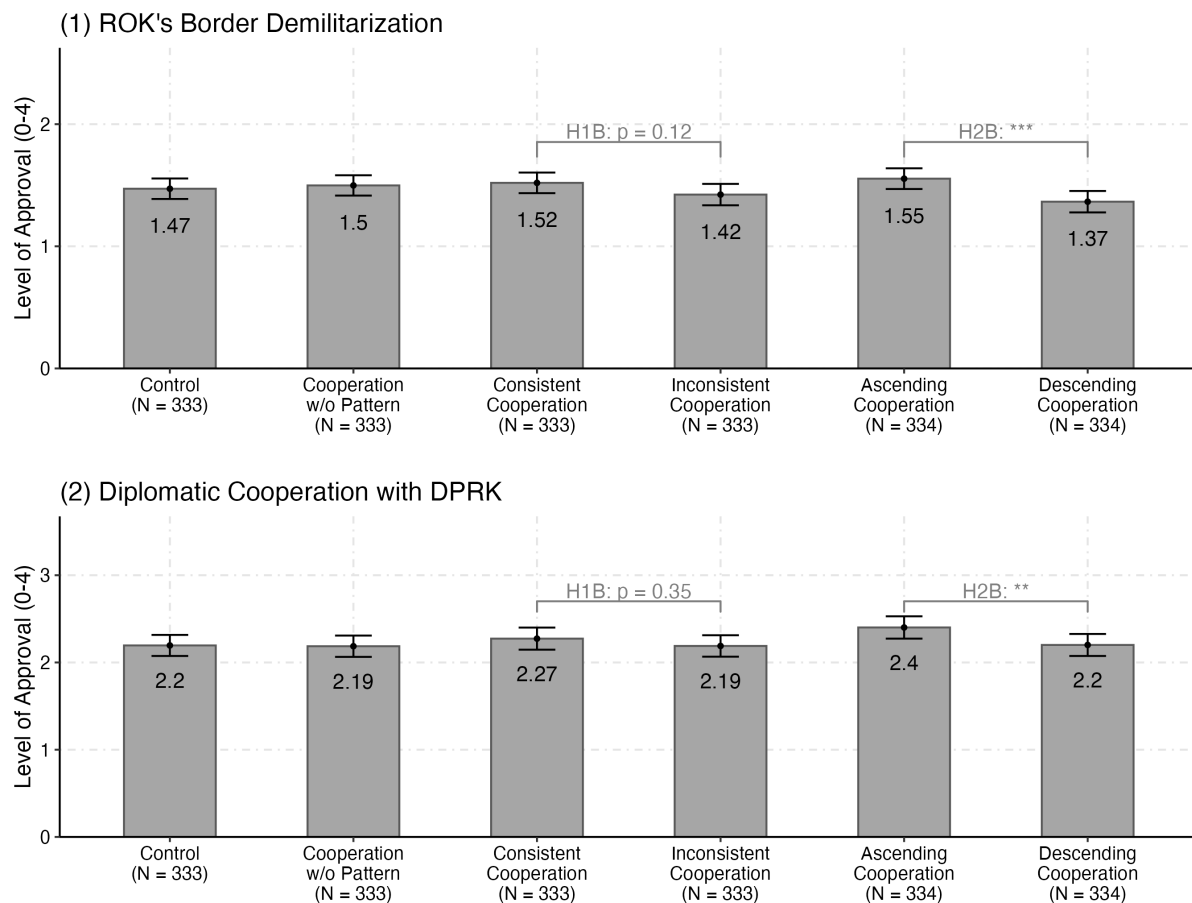
Note: Error bars represent 95 percent confidence intervals. ***: $p < 0.01$; **: $p < 0.05$; *: $p < 0.1$.

cooperation, consistent cooperation significantly increases expectations, with a mean difference of 0.19 ($p = 0.02$), while inconsistent cooperation produces a negative but statistically insignificant effect, with a mean difference of -0.10 ($p = 0.23$).

Turning to [H2A](#), which predicts that descending cooperation should generate less optimistic views regarding North Korea's future cooperation than ascending cooperation, the fifth and sixth columns in the figure support this expectation. For both outcome variables, respondents exposed to an ascending cooperative pattern report higher mean scores than those in the descending condition — 1.53 versus 1.21 for military disengagement and 1.54 versus 1.2 for diplomatic cooperation — with both differences statistically significant at the 0.01 level. The gap appears to be driven primarily by the negative effect of declining cooperation, which significantly reduces cooperative expectations relative to the control group with a mean difference of -0.24 ($p < 0.01$) for military disengagement and -0.26 ($p < 0.01$) for diplomatic cooperation, while ascending

cooperation produces only a modest increase with a mean difference of 0.06 ($p = 0.46$) for military disengagement and 0.10 ($p = 0.24$) for diplomatic cooperation.

Figure 7: Patterns of Cooperation and Public Attitudes Toward Rapprochement with North Korea



Note: Error bars represent 95 percent confidence intervals. ***: $p < 0.01$; **: $p < 0.05$; *: $p < 0.1$.

Next, we examine how the patterns of cooperation shape South Korean respondents' approval of conciliatory policies toward North Korea, as displayed in Figure 7. We use two outcome variables measured on a 5-point Likert scale (0–4): (1) approval of South Korea's border demilitarization and (2) approval of diplomatic cooperation with North Korea. As shown in the first column of the graphs, respondents in the control condition report mean scores of 1.47 and 2.20, respectively. This suggests a divergence in baseline attitudes: South Koreans are relatively skeptical of border demilitarization but more receptive to diplomatic cooperation. Additionally, exposure to a description of North Korea's 2018 cooperation without any temporal pattern did not significantly alter approval for either outcome with a mean difference of 0.03 ($p = 0.65$) for border demilitarization and -0.01 ($p = 0.92$) for diplomatic cooperation, consistent with the

pattern previously observed from the expectation outcomes.

According to [H1B](#), we expect that inconsistent cooperation should generate lower public approval for conciliatory policies than consistent cooperation. The third and fourth columns of [Figure 7](#) provide suggestive evidence supporting this hypothesis. For border demilitarization, respondents in the consistent condition (1.52) report a slightly higher mean score than those in the inconsistent condition (1.42), but this difference does not reach conventional statistical significance ($p = 0.12$). Similarly, for diplomatic cooperation, the mean scores are nearly identical between the two conditions (2.27 versus 2.19), with no statistically significant difference ($p = 0.35$). The absence of a significant effect here suggests that the distinction between consistent and inconsistent cooperation — while meaningful for shaping expectations of adversary behavior — may be insufficient to alter support for concrete conciliatory policies.

One possible explanation is that the treatment was strong enough to revise respondents' perceptions of North Korea's future intentions but not strong enough to change their underlying policy preferences. This interpretation is consistent with the manipulation check reported in Appendix 2.3, which indicates that the treatment of inconsistent cooperation was the least effective at altering respondents' perceptions of the 2018 diplomatic episode. More broadly, attitudes toward reconciliation may be anchored in stable political predispositions and prior beliefs that are less responsive to a single informational intervention than expectations about future behavior. As a result, respondents may update their assessments of North Korea's likely conduct without correspondingly revising their support for specific conciliatory policies.

Regarding [H2B](#), which predicts that descending cooperation should reduce policy approval compared to ascending cooperation, the fifth and sixth columns in the graphs offer supportive evidence. For both outcome variables, respondents in the ascending condition report higher mean scores than those in the descending condition — 1.55 versus 1.37 for border demilitarization and 2.40 versus 2.20 for diplomatic cooperation — with the difference reaching statistical significance at the 0.05 level. Breaking down these differences reveals a mixed pattern across the two outcome variables. For border demilitarization, ascending cooperation produces a positive but insignificant increase with a mean difference of 0.08 ($p = 0.17$), while descending cooperation reduces approval with a mean difference of -0.11 ($p = 0.09$). For diplomatic co-

operation, however, the gap is driven primarily by the positive effect of ascending cooperation, which significantly increases policy approval relative to the control group with a mean difference of 0.21 ($p = 0.02$), while descending cooperation produces only a negligible and statistically insignificant change with a mean difference of -0.01 ($p = 0.95$).

We conclude the empirical analysis with robustness checks. As shown in Appendix 2.3, our manipulation check confirms that the treatment vignettes successfully induced the intended perceptions of cooperation trajectories. According to the results, respondents assigned to each patterned condition are significantly more likely to characterize North Korea’s 2018 cooperation in line with their assigned vignette, providing confidence that the treatment effects reported above reflect respondents’ active engagement with the cooperation patterns rather than noise. Additionally, the balance check reported in Appendix 2.4 indicates no statistically significant differences in demographic characteristics across the experimental groups.

7. Conclusion

This paper examined how the trajectory of cooperative signals during rapprochement shapes public expectations about an adversary’s intentions and support for reciprocal concessions. We argue that cooperation trajectories matter because even broadly positive episodes of rapprochement can be undermined by downturns in cooperation, which reinforce preexisting beliefs that the adversary is inherently hostile. As a result, individuals who observe cooperation that is punctuated by setbacks or deteriorates over time are likely to expect less cooperation in the future and to express greater resistance to peace initiatives.

Two preregistered survey experiments provided consistent support for these arguments. In Study I, American respondents exposed to inconsistent or descending trajectories expressed lower approval of peace relative to their relevant counterfactuals: consistent and ascending trajectories, respectively. Study II replicated the expectation results in a historically grounded setting and provided more qualified evidence for policy attitudes, especially along the directional dimension. South Korean respondents exposed to inconsistent or descending portrayals of North Korea’s behavior during the 2018 inter-Korean rapprochement reported lower expecta-

tions of future military and diplomatic cooperation and weaker support for conciliatory policies. Together, these findings demonstrate that the temporal pattern of cooperation shapes public support for rapprochement beyond the mere aggregate level of cooperative behavior.

More broadly, these findings contribute to two related strands of IR scholarship. The costly-signaling literature argues that bold and dramatic gestures are particularly effective at conveying sincere intentions under conditions of deep distrust (Kydd, 2005; Kertzer et al., 2020; Yoder & Haynes, 2021; S. J. Kim, 2021; Mattes & Weeks, 2022). A parallel tradition emphasizes that trust emerges not from any single act, but from cooperation that is sustained across repeated interactions over time (Osgood, 1962; Axelrod & Keohane, 1985; Hoffman, 2006; Crescenzi, 2018). Our findings suggest that neither perspective alone is sufficient. Rather, public support for rapprochement depends not only on the credibility of individual signals, but also on the trajectory through which cooperative behavior unfolds over time. Historical cases illustrate this dynamic. Egyptian President Sadat's visit to Jerusalem was among the most dramatic unilateral gestures in modern diplomacy, yet societal hostility persisted long after the Camp David Accords. Likewise, the unprecedented inter-Korean summits of 2018 generated considerable public optimism that ultimately proved difficult to sustain. Our results suggest that even highly credible and costly gestures may fail to produce durable public support for peace when they are followed by declining cooperation or embedded within an inconsistent pattern of engagement. In such cases, citizens may interpret cooperation as a temporary deviation from, rather than evidence against, an adversary's underlying hostility.

Moreover, our findings carry policy implications. First, leaders pursuing rapprochement should attend not only to the magnitude of individual cooperative gestures but to the sequencing through which they are presented to domestic audiences. A strategy that begins with smaller confidence-building measures and escalates toward more substantive commitments is more likely to sustain public optimism than a strategy that opens with dramatic overtures. Second, interruptions to cooperative momentum are costly in a way that symmetric recovery cannot easily offset. There is a tension at the heart of peace negotiations: the back-and-forth bargaining that characterizes real diplomatic processes may be strategically necessary at the negotiating table but corrosive to public support when made visible. Leaders should therefore consider shielding

the iterative mechanics of negotiation from public view until a durable agreement is in place, or framing temporary setbacks within a broader narrative of forward momentum, treating cooperative continuity as a strategic imperative. Third, leaders who communicate a positive and forward-looking narrative of rapprochement — one that makes the arc of progress visible and legible to domestic audiences — are better positioned to sustain public support for conciliatory policy. Our evidence that visual and narrative representations of cooperation trajectories shape public expectations suggests that framing the peace process as an unfolding story of deepening commitment, rather than a series of discrete diplomatic events, is an important tool for shaping public opinion.

This study has several limitations that future research should address. The first limitation concerns our treatment of retrospective evaluation as a summary judgment formed at a single point in time. In practice, citizens' cumulative assessment of a cooperation trajectory may itself be shaped by how they evaluated each individual cooperative gesture as it occurred, with early impressions conditioning the interpretation of subsequent signals in ways our design cannot capture. Future research that tracks public evaluations dynamically across the unfolding of rapprochement would therefore enrich our understanding of how the accumulation of those real-time evaluations determines whether elite-level rapprochement ultimately generates the societal conciliation that defines warm peace. Second, both empirical studies in this paper rely on experimental methods that present cooperation trajectories in a simplified form, abstracting away from complex political contexts. Specifically, the sparse information environment of Study I may cause us to overestimate trajectory effects relative to more naturalistic settings, and although Study II addresses this through a selective-history design, prior knowledge of the 2018 events may have attenuated treatment effects. Connecting these micro-level findings to observational data on real rapprochement processes would therefore be a next step, helping to establish whether the trajectory mechanism identified here operates with comparable force in settings where cooperation unfolds amid the full complexity of domestic politics, strategic uncertainty, and media intermediation.

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